

Section-A (25 Marks)

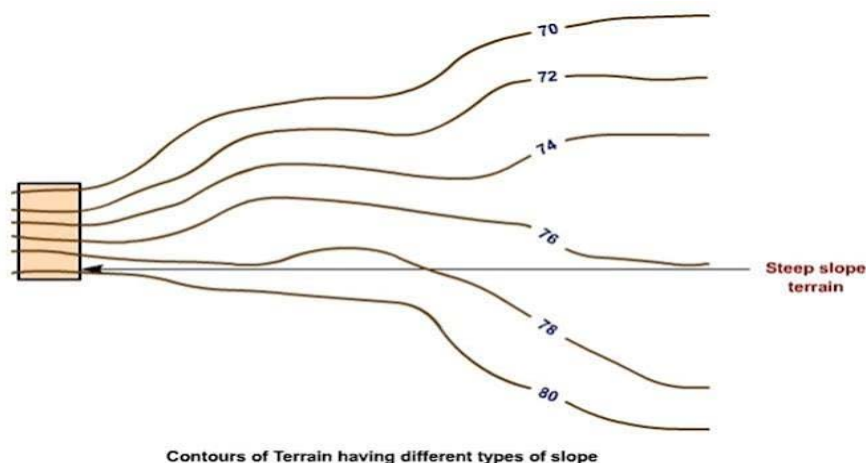
- 1. Write down the characteristics of contour lines. Why is engineering survey important while preparing detailed project report of a water supply project? Explain. [4+6=10]**

Answer:

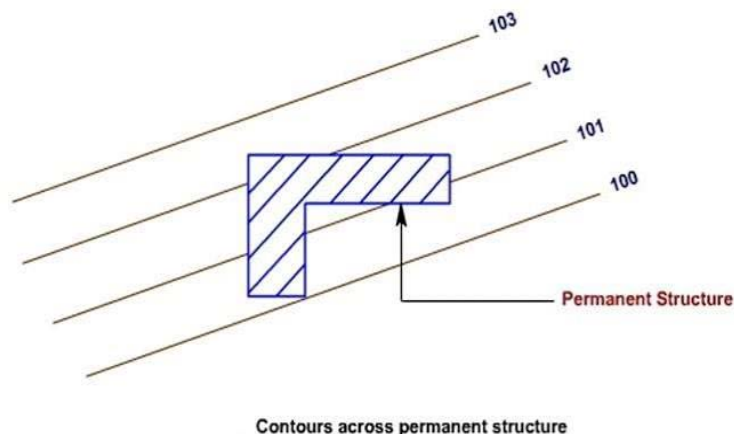
A contour is defined as an imaginary line of constant elevation on the ground surface. A line joining the points of equal elevations is called a contour line. A contour map consists of contour lines with its elevation to keep the country map simple and easy to read.

The characteristics of contour lines are listed below:

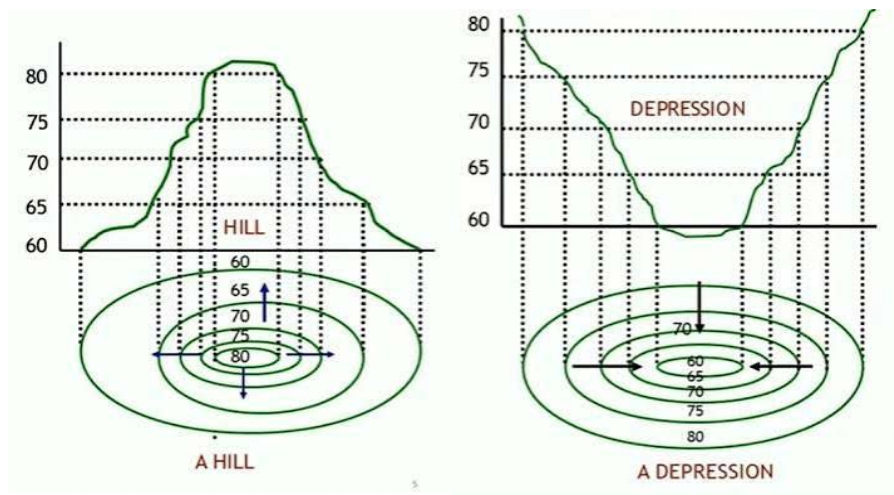
- The vertical distances between any two control lines are assumed to be uniform.
- The horizontal distance between any two contour lines indicates the amount of slope and varies inversely on the amount of slope.



- Contours do not pass through permanent structures such as buildings.



- Contours of different elevation cannot cross each other (caves and overhanging cliffs are the exceptions.)
- A close contour line on a map represents either depression or hill. A set of contours with higher values inside represent a hill whereas the lower value inside depicts a depression.



Engineering survey is one of the most important operations for the preparation of detailed project report of any engineering project. The survey helps to obtain accurate field data required for planning, design, construction and cost estimation of the project. The project faces technical, economic and many other challenges if proper engineering survey is not performed.

Importance of engineering survey while preparing DPR of water supply projects are explained in points below:

i. Identification of source:

- Engineering survey is carried out to determine the suitable source and groundwater sources.
- These surveys help to assess the availability, suitability, reliability of source for water supply projects.

ii. Determination of ground levels:

- Contour maps and elevation data are obtained from engineering survey.
- They are essential for determination of storage capacity of reservoirs, location of reservoirs, pumping requirements, designing of gravity flow systems and so on.

iii. Route selection and pipeline alignment:

- Engineering surveys locate the obstruction such as rivers, hills, buildings etc. that are helpful in selection of most economical and feasible route and pipeline alignment.
- Survey also determines elevations, slopes, and terrain features to select the optimal pipeline alignment.
- Proper engineering survey helps in avoiding costly obstacles and ensuring gravity flow where possible.

iv. Design of hydraulic structures:

- Data obtained from surveys help to calculate pipeline diameter, required pressure, and flow rate needed to transport water from source to treatment facility to customers.
- Design of treatment plants, pumps and other hydraulic structures also utilize the data and observations obtained from engineering surveys.

v. Accurate quantity and cost estimation:

- Accurate measurements from different engineering surveys enable accurate quantity estimation of materials and allows to estimate project cost for project approval.
- Survey is also equally helpful for realistic project budgeting and financial planning.

vi. Assessment of feasibility of construction:

- Surveys help to identify the type of terrain, unstable slopes and accessibility.
- It helps to conclude the feasibility and economic viability of the construction project

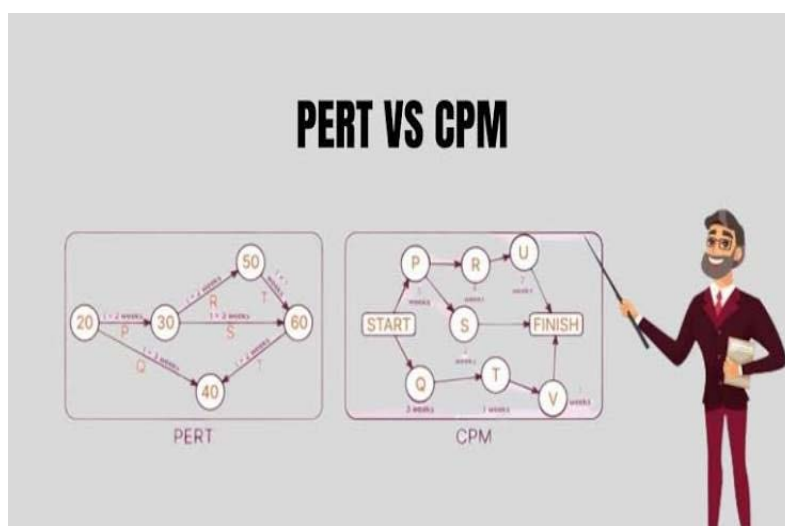
2. Differentiate between CPM and PERT. Describe the stages of a project at which cost control is affected. [5+5=10]

Answer:

CPM (Critical Path Method) and PERT (Program Evaluation and Review Technique) are the two techniques of project management useful for planning, scheduling and controlling projects. Both CPM and PERT share same approach for constructing the project network and for determining the critical path of the network.

The **differences between CPM and PERT** are mentioned below:

CPM	PERT
1. Time estimates in CPM method are deterministic. Hence, single time estimate is used for activity duration.	1. Time estimates in PERT method are probabilistic with uncertainty in activity duration. Hence, three-time estimates i.e. pessimistic time, optimistic time and most-likely time estimates are used.
2. CPM is activity-oriented network diagram.	2. PERT is event-oriented network diagram.
3. This method is more suitable for simple and repetitive projects like construction.	3. This method is more suitable for new and complex projects like research and developmental projects.
4. CPM is easy to maintain.	4. PERT is costly to maintain.
5. It is focused on time cost trade off.	5. It is focused on time only, no cost.



3. Why is the 28 days compressive strength considered the standard for determining the grade of concrete? And, how do curing conditions affect the strength? Clarify. [5+5=10]

Answer:

28-day compressive strength of concrete is the maximum compressive stress that a standard concrete specimen can withstand after being cured for 28 days, expressed in MPa (N/mm²). It is treated as standard for determining the grade of concrete. for example: Grade M20 concrete means that a standard concrete cube attains a characteristic compressive strength of 20 MPa (20 N/mm²) at 28 days of curing.

28 days compressive strength is considered the standard for determining the grade of concrete because:

- About 65–75% of the ultimate strength is achieved in the first 7 days and about 90% or more by 28 days under normal curing conditions.
- Most design codes and standards use 28-day compressive strength as the basis for concrete classification and structural design.

- iii. Gaining full ultimate strength may take long period of time which is impractical for the construction projects. Hence, 28 days period offers reasonable balance between accuracy and construction schedule.
- iv. Allows engineers to compare different concrete mixes and ensure consistent quality of the projects.
- v. It helps to consider normal variation in materials, curing, workmanship by using standard age and a statistical strength criterion.

Curing of concrete is the process of keeping freshly placed concrete moist for a certain period at specific temperature so that it can gain the required strength and durability. Curing conditions controls the hydration of cement which affect concrete strength. Longer curing allows more hydration, so strength keeps developing for a longer period of time.

Typical strength development:

7 days: About 65–75% of 28-day strength.

28 days: About 90% or more of ultimate strength.

Beyond 28 days: Strength continues to increase gradually.



Curing conditions affect the strength of concrete by:

- i. Providing the adequate amount of water required for hydration of cement.
- ii. Reducing shrinkage cracks.
- iii. Increasing compressive strength and durability.
- iv. Producing dense concrete.
- v. Increasing the resistance against chemical action and weathering.

4. How is water demand managed in the Kathmandu valley? And, what challenges and strategies are associated with meeting the growing needs of its population? Discuss. [4+6=10]

Answer:

Water demand is the total volume of water required to fulfill various human activities and sustain ecological requirements. The daily water demand for Kathmandu Valley is more than 360 million liters per day, whereas the supply is around 90 million liters per day in dry season. It indicates that water demand has grown faster than reliable water supply.

Water demand in the Kathmandu Valley is managed through a combination of surface water supply, groundwater extraction, demand management measures, and augmentation projects.

i. Utility supply and distribution:

Kathmandu Upatyaka Khanepani Limited (KUKL) is responsible for distributing the drinking water in Kathmandu Valley. Water from rivers, springs, reservoirs, and groundwater sources are tapped and supplied through a piped distribution network. Limited supply of water from the source has resulted to intermittent supply of water on a rotational schedule.

ii. Groundwater extraction:

Groundwater from deep and shallow tube wells is used to supplement municipal supply. Many households, institutions, and industries operate private wells to meet their water demand. Kathmandu Valley Water Supply Management Board (KVWSMB) has prepared groundwater management policy measures to restrict extraction of groundwater in danger zones, require registration of tube wells, and license commercial extraction.

iii. Alternative measures:

Water tankers and private suppliers also supply water during shortages. Community and household storage tanks are commonly used to manage water needs. Measures such as rainwater harvesting can be used to recycle and reuse the water reducing the wastages of water.

iv. Demand management measures:

Water meters and tariff systems are incorporated in the distribution pipelines at the individual households for efficient use of water. Leak detection, repair works, help reduce losses and non-revenue water (NRW) in the distribution network.

v. Future expansion:

Expansion of water treatment and distribution infrastructure.

Main challenges:

i. Rapid Population Growth:

Water demand has significantly increased due to increasing population and urbanization. However, limited water supply from the limited natural sources has resulted in the huge gap between water supply and demand causing scarcity of water.

ii. Over-Extraction of Groundwater

To meet the increasing water demand, large amount of water of drawn from the groundwater. However, excessive pumping has caused declining groundwater levels leading to depletion of groundwater resources.

iii. Aging Infrastructure and Leakage

Old pipelines suffer leakages and frequent breakdowns. Significant portion of supplied water is lost before reaching consumers. This causes interruption in water supply to the consumers leading to water scarcity at dry regions.

iv. Water Pollution

Rivers and groundwater sources are contaminated by dumping of untreated domestic sewage, industrial waste, and solid wastes directly on these natural resources. Additional treatment is required to make water safe for drinking.

v. Climate Change Impacts

Water scarcity has become a common challenge due to changes in rainfall patterns and prolonged dry periods as a result of climate change and global warming. Increased risk of droughts and water shortages gives rise to greater problems such as epidemics and pollution.

vi. High Non-Revenue Water

Water losses due to leakage, illegal connections, and inaccurate meter readings reduce water supply system efficiency.

Strategies associated with meeting the water demand of growing population:

i. Alternative methods of water conservation:

Techniques such as groundwater harvesting, grey-water managements are should be adopted to recycle and reuse the greywater or surface runoff for cattle, vegetation growth, toilet flushing and so on. These practices reduce the fresh water demand, preventing the wastages of fresh water for minor use and reduces the dependency on municipal water supply.

ii. Groundwater Recharge and Conservation:

Construction of recharge ponds in the urban area helps in efficient groundwater recharge as the surface runoff in this area are larger due to larger built-up surface. Construction of parks and less concrete built-up surface, facilitates the conservation and recharge of groundwater levels.

iii. Improvement of infrastructures:

Maintenance of old pipelines, pumps and components of water supply systems also help to tackle the problem of water scarcity. Increasing the capacity of water treatment and supply systems also help to meet the increasing water demand of present and also of the future population.

iv. Watershed protection:

Natural resources such as forests, vegetation, rivers, lakes and catchment area must be protected and preserved so that these resources naturally replenish the extracted groundwater through infiltration and recharge.

5. What are the ideal conditions of fixing of river intake? Illustrate the general procedure for design of transmission pipes in gravity water supply system. [3+7 =10]

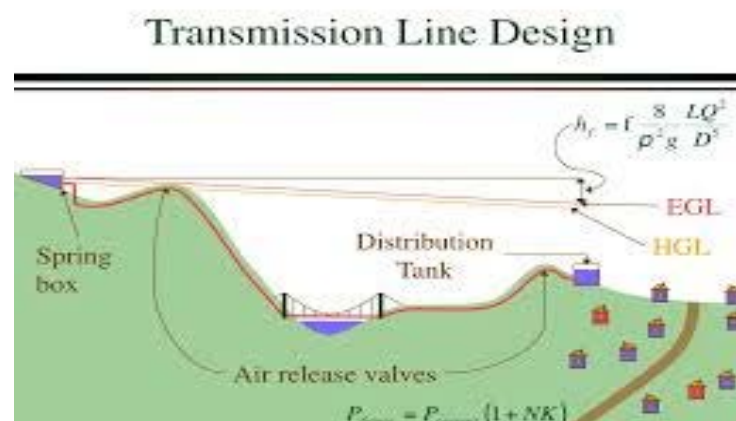
Answer:

River intake is the hydraulic structure constructed at or near the river source to draw the required amount of discharge from the source. The main function of intake is to collect water from the source and discharge the collected water to the treatment plant directly or by means of pump.

The ideal conditions of fixing of river intake are described as follows:

- i. The site should be near the treatment plant as possible so that the cost of conveying water will be less due to shorter conveyance line.
- ii. Intake should be located in such a place where sufficient quantity of water is available during the dry season too.
- iii. It should not be located near the navigation channel due to possibility of pollution by the discharge of refuse from ships and boats.
- iv. Intake sites should be easily accessible during floods but should not get flooded.
- v. The intake should not be located on the meandering curves however if it has to be constructed on the meander then it should be made on the outer bank so that what is available all the time.

General procedure for the design of transmission pipes in gravity water supply system is shown and described below:



i. Data collection and selection of design period:

- At first, survey is carried out to determine the discharge available in the source, population demand, topography, route length, and levels of intake, pipe alignment, and reservoir.
- Appropriate design period for the project must be selected to estimate the future population by the help of population forecasting methods.

ii. Estimation of water demand:

- Per capita water demand is calculated.
- Determine the total design discharge:

$$Q = P \times q$$

Where:

Q = Total water demand (L/day)

P = Design population

q = Per capita demand (L/capita/day)

iii. Selection and layout of route:

Topographic and engineering surveys are carried out to determine the pipe alignment, ground profile, and elevations of source and destination.

iv. Calculation of available head:

Determine the difference in elevation between the water source and the receiving reservoir.

$$H = Z_1 - Z_2$$

Where:

H = Available head (m)

Z_1 = Elevation of source water level

Z_2 = Elevation of receiving reservoir water level

v. Fix the design flow:

The maximum daily or peak hourly flow that the transmission main should carry is estimated.

vi. Selection of pipe material:

Suitable pipe material for example: DI, MS, HDPE, PVC, etc. based on pressure, durability, cost, and site conditions are selected.

vii. Determination of pipe diameter:

A suitable flow velocity, usually 0.6–3.0 m/s is assumed.

Compute pipe diameter using:

$$Q = \frac{\pi D^2}{4} V$$

Where:

Q = Discharge (m³/s)

D = Diameter (m)

V = Velocity (m/s)

viii. Head loss calculation:

Calculate friction losses using the Hazen–William's equation:

$$h_f = \frac{10.67 L Q^{1.852}}{C^{1.852} D^{4.87}}$$

Where:

h_f = Friction head loss (m)

L = Length of pipe (m)

Q = Discharge (m^3/s)

D = Diameter (m)

C = Hazen-William's coefficient

Also, minor losses due to bends, valves, entrances, and exits are also considered.

ix. Check HGL and pressure conditions:

Hydraulic Adequacy is checked by:

$$H \geq h_f + h_m$$

Where:

H = Available head

h_f = Friction loss

h_m = Minor losses

If not satisfied, pipe diameter is increased and calculations are repeated. The positive pressure must be ensured throughout the pipeline. Excessive pressures that may damage the pipe must be avoided.

x. Design of appurtenances:

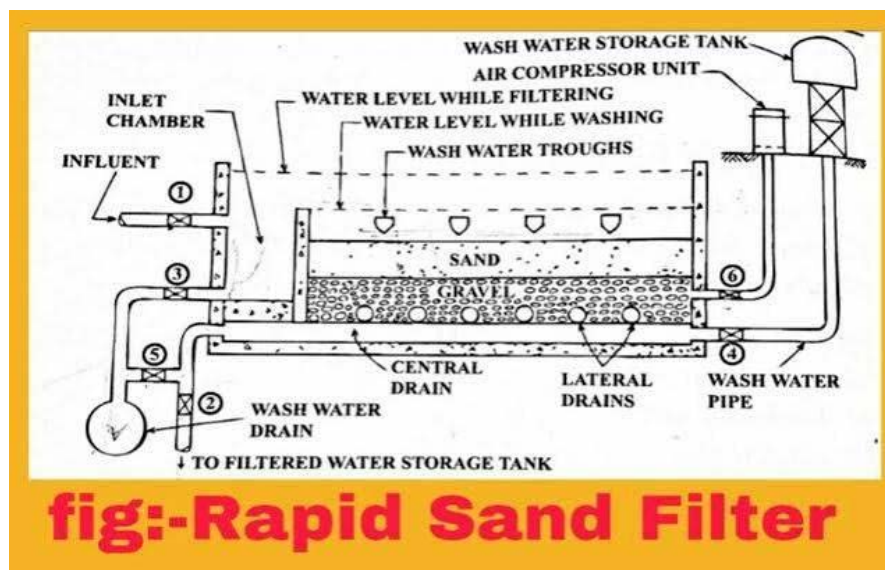
Appurtenances such as valves, pumps, bends, fittings must be appropriately designed. Provide air valves at high points, washout valves at low points, isolation valves at suitable intervals.

Section-B (25 Marks)

6. Explain the problems encountered during the operation of rapid sand filters in water treatment and suggest how these problems can be controlled. [5+5=10]

Answer:

A rapid sand filter is a high-rate biological water filtration system used in municipal treatment plants to remove suspended and dissolved solids from the water. This filter is highly effective in color or turbidity removal but less effective in removal of bacterial load in comparison to slow sand filter.



The **problems encountered during the operation of rapid sand filters** in water treatment are as follows:

i. Cracking or damaging of filter bed:

Drying of filter beds, shrinkage and uneven filtration causes cracks in the bed of rapid sand filter. Cracked filter bed allow water to pass through it without adequate filtration.

ii. Excessive head loss:

Accumulation of suspended solids and clogs in the filter media increase the resistance to flow which increases head loss.

iii. Algal growth:

Algae may develop on filter surfaces exposed to sunlight which results in frequent clogging and odor nuisance.

iv. Formation of mud balls:

Small balls of mud and organic matter form within the filter bed due to inadequate backwashing. These mud balls create channels causing frequent clogging and reduce filtration efficiency.

v. Air binding:

The accumulation of air within the filter bed during excessive head loss creates negative pressure. This blocks pore spaces and reduces filter performance.

These **problems can be encountered by** the following ways:

- i. Proper and adequate backwashing operations must be carried out at a frequent interval.
- ii. Constant water level must be maintained about the filter bed to prevent cracking of filter due to uneven filtration, drying and air binding.
- iii. Air release arrangements must be provided wherever necessary.
- iv. Pretreatment of water such as sedimentation with coagulation should be carried out before passing the water to the filter.
- v. Proper functional under-drain systems must be incorporated for proper backwashing operations.
- vi. Timely cleaning and replacement of filter bed of the rapid sand filter must be carried out.

7. List the treatment methods used for water softening and describe the lime soda process in detail. [2+8=10]

Answer:

The removal or reduction of permanent and temporary hardness of water is known as water softening. It is the process of removal of calcium and magnesium ion from the water. The process includes:

- i. Removal of temporary hardness
- ii. Removal of permanent hardness

i. Removal of temporary hardness:

The temporary hardness of water is due to the presence of bicarbonates of calcium and magnesium. It can be removed by two methods: boiling method and addition of lime method.

a. Boiling method:

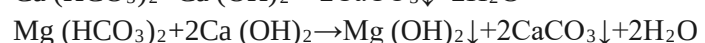
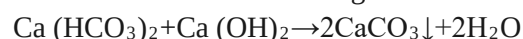
When the water is boiled having the temporary hardness the following chemical reaction takes place.



The calcium and magnesium carbonates are insoluble in water and are removed by sedimentation tanks. However, this is very costly process and not possible to use in water works.

b. Addition of lime:

When the lime is added to the water following reaction takes place:



The insoluble precipitates MgCO_3 and CaCO_3 are removed by sedimentation.

ii. Removal of permanent hardness:

The permanent hardness of water can be removed by following process:

- soda lime method
- zeolite method or base exchanger process
- ionization method or demineralization process

Lime soda method is explained in detail below:

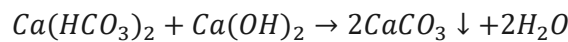
The Lime-Soda Process is a chemical water-softening method used to remove both temporary hardness and permanent hardness from water by adding calculated amounts of lime [$\text{Ca}(\text{OH})_2$] and soda ash [Na_2CO_3].

Principle:

Hardness in water is mainly caused by dissolved salts of calcium (Ca^{2+}) and magnesium (Mg^{2+}). Lime and soda react with these salts to form insoluble precipitates of (CaCO_3) and [$\text{Mg}(\text{OH})_2$]. These precipitates settle out and are removed by sedimentation and filtration.

A. Removal of Temporary Hardness

➤ Calcium Bicarbonate

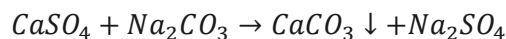


➤ Magnesium Bicarbonate

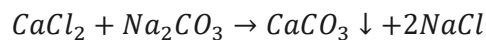


B. Removal of Permanent Hardness

➤ Calcium Sulphate

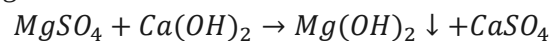


➤ Calcium Chloride

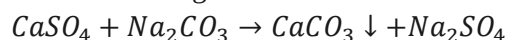


➤ Magnesium Sulphate

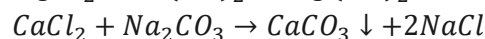
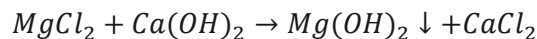
lime converts magnesium salt into calcium salt:



Then soda removes the resulting calcium salt:



➤ Magnesium Chloride



Procedure:

1. Chemical Addition

Calculated quantities of lime and soda ash are added to raw water.

2. Rapid Mixing

Chemicals are thoroughly mixed with water to ensure complete reaction.

3. Flocculation

The precipitates of CaCO_3 and $\text{Mg}(\text{OH})_2$ form larger flocs.

4. Sedimentation

Flocs settle at the bottom of the sedimentation tank.

5. Filtration

The clarified water is filtered, usually through rapid sand filters.

6. Disinfection

Chlorination is often carried out before distribution.

Advantages:

- i. Less quantity of coagulant is required in lime soda process.
- ii. Removal of iron and manganese to some extent.
- iii. The process is most suitable for turbid and acidic waters.

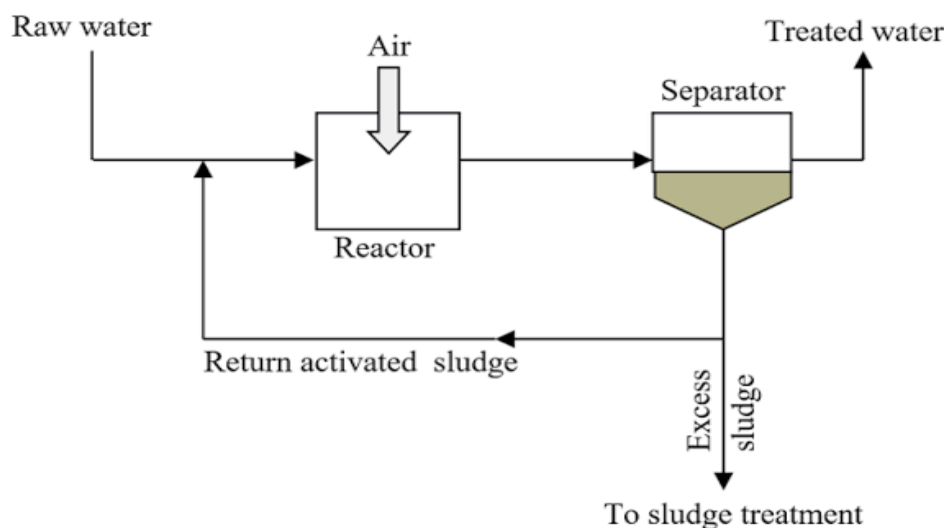
Disadvantages:

- i. Process requires skilled supervision for its successful working.
- ii. Large quantity of sludge is formed during this process which needs it be disposed off.
- iii. A thick layer of calcium carbonate will be deposited in filtering media.

8. What are the main stages and components involved in the activated sludge process for wastewater treatment? And, what are the important parameters that needed to be monitored? Explain. [7+3=10]

Answer:

Activated sludge process is a biological method of wastewater treatment that utilizes a mixture of microorganisms such as bacteria, fungi, protozoa and rotifers maintained in suspension, in an aerobic environment to metabolize organic matter to produce secondary wastewater effluent of acceptable quality. It has removal efficiencies in the range of 95-98% than trickling filters of 80-85%.



The main stages involved in activated sludge process is described below:

i. Physiochemical adsorption:

The fine suspended particles of the sewage combine with bacteria to form a sub layer of bacterial film at the surface. The film attracts finely divided colloidal and dissolved organic matter and result in formation of flocs. The flocs get settled at the bottom of sedimentation tanks and are removed.

ii. Oxidation:

Large flocs formed act as a vehicle for aerobic bacteria that oxidizes the carbonaceous and organic matter of wastewater by biochemical action. Settled microorganisms are removed from the bottom of settling tank are and are returned to aeration tank to metabolize additional organic matter. Balance between food and mass of the organisms, also known as sludge loading ratio must be maintained.

Components involved in activated sludge process:

- i. A reactor or aeration tank in which the microorganisms responsible for treatment are kept in suspension and are aerated.

- ii. Final clarifier or secondary settling tank for liquid solid separation by gravity.
- iii. A recycle system for returning some portion of activated sludge from clarifier to reactor.
- iv. Discharging of settled effluent.
- v. Wasting of excess biomass.

Working:

- A primary clarifier may be introduced to remove part of suspended solid present in the influent and this reduces the organic load to the activated sludge system.
- The biological aerator is filled with a mixer of activated sludge and influent known as mixed liquor. It is necessary to maintain certain liquid suspended solid (MLSS) in the aerated tank which maintain good removal efficiency.
- The aerator equipment transfers oxygen into the reactor for the oxidation of organic matter.
- The continuous introduction of new influent results in a continuous discharge of mixed liquor to the secondary settling tank where separation of solid and liquid takes place.
- The liquid leaves the system as a treated effluent whereas some part of this sludge is recirculated to aeration tank called as return sludge and rest is taken for anaerobic digestion.

The important parameters that need to be monitored:

i. Dissolved oxygen:

Amount of oxygen available in the aeration tank should be constantly monitored to maintain aerobic condition in the tank. DO of 1.5–3.0 mg/L, usually around 2 mg/L is maintained in the activated sludge process.

ii. Mixed liquor suspended solids:

The concentration of suspended solids (microorganisms and organic matter) in the aeration tank should also be properly monitored in this activated sludge process. MLSS of 2,000–4,000 mg/L is maintained.

iii. Food-to-microorganisms ratio:

$$F/M = \frac{\text{BOD load applied per day}}{\text{Mass of microorganisms in aeration tank}}$$

High F/M leads to formation of young sludge, unstable operation whereas low F/M results in formation of old sludge and endogenous respiration.

Typical Range: 0.2–0.5 kg BOD/kg MLVSS·day

iv. Sludge volume index (SVI):

$$SVI = \frac{\text{Settled sludge volume (mL/L)}}{\text{MLSS (mg/L)}} \times 1000$$

Typical Range: 80–150 mL/g

Interpretation:

SVI < 80 mL/g: Dense sludge

SVI = 80–150 mL/g: Good settling

SVI > 150 mL/g: Bulking sludge

v. Sludge retention time:

Average time microorganisms remain in the system.

$$SRT = \frac{\text{Mass of solids in system}}{\text{Mass of solids wasted per day}}$$

Typical Range: 5–15 days

9. Design a combined sewer section for an urban area with the following information. [10]

Area=10 ha

Population density=400 persons/ha

Per capita water demand=10 L/day

Rainfall intensity=50 mm/hour

Assume other relevant parameters.

Answer:

Given Data:

Area, $A = 10\text{ha} = 100,000\text{ m}^2$

Population density = 400 persons/ha

Population, $P = 10 \times 400 = 4000\text{persons}$

Per capita water demand = 10 L/day

Rainfall intensity, $I = 50\text{mm/hr}$

Runoff coefficient, $C = 0.75$

Manning's coefficient, $n = 0.013$

Sewer flowing full

Assume, Bed slope, $S = 1/500 = 0.002$

Calculate Sanitary Sewage Flow:

Average water demand:

$$Q_w = 4000 \times 10 = 40000\text{ L/day}$$

Assuming 80% of water supply reaches the sewer:

$$Q_s = 0.8 \times 40000 = 32000\text{ L/day}$$

$$Q_s = \frac{32000}{86400}$$

$$Q_s = 0.370\text{ L/s}$$

Applying peak factor = 3:

$$Q_{sp} = 3 \times 0.370 = 1.11\text{ L/s}$$

$$Q_{sp} = 0.00111\text{ m}^3/\text{s}$$

Calculate Storm Water Flow:

Using Rational Formula:

$$Q_r = \frac{CIA}{360}$$

Where:

$$C = 0.75$$

$$I = 50\text{ mm/hr}$$

$$A = 10\text{ ha}$$

$$Q_r = \frac{0.75 \times 50 \times 10}{360}$$

$$Q_r = 1.042\text{ m}^3/\text{s}$$

Total Combined Sewer Discharge:

$$Q = Q_r + Q_{sp}$$

$$Q = 1.042 + 0.00111$$

$$Q = 1.043\text{ m}^3/\text{s}$$

Design Sewer Diameter:

Using Manning's Formula for full flow:

$$Q = \frac{1}{n} AR^{2/3} S^{1/2}$$

For a circular sewer flowing full:

$$A = \frac{\pi D^2}{4}$$

$$R = \frac{D}{4}$$

Substituting:

$$Q = \frac{1}{0.013} \left(\frac{\pi D^2}{4} \right) \left(\frac{D}{4} \right)^{2/3} (0.002)^{1/2}$$

$$1.043 = 1.697 D^{8/3}$$

$$D^{8/3} = 0.6146$$

$$D = (0.6146)^{3/8}$$

$$D \approx 0.833 \text{ m}$$

Provide the next standard size:

$$\boxed{D = 900 \text{ mm}}$$

Check Velocity:

For $D = 0.9 \text{ m}$:

$$A = \frac{\pi(0.9)^2}{4} = 0.636 \text{ m}^2$$

$$V = \frac{Q}{A} = \frac{1.053}{0.636}$$

$$V = 1.66 \text{ m/s}$$

Since $0.6 < V < 3.0 \text{ m/s}$, the design is satisfactory.

Hence, a **900 mm diameter circular combined sewer** laid at a slope of **1 in 500** is suitable for the given urban area.

10. What are the types of sewage filters? Explain the biological process involved in trickling filter. Discuss the merits and demerits of trickling filters in the sewage treatment. [2+5+3=10]

Answer:

Sewage filters are the filter units that are used for biological treatment of wastewater, consisting of open beds of filtering media over which the affluent from primary settling tank is passed intermittently.

There are **three types of sewage filters**. They are:

i. Intermittent sand filter:

- It is a specially prepared bed of sand on which effluents from primary treatment may be applied intermittently by using troughs or perforated pipes.

ii. Contact bed:

- It is a biological sewage treatment process consisting of a watertight tank filled with coarse filter media, such as broken stones or bricks.

iii. Trickling filter:

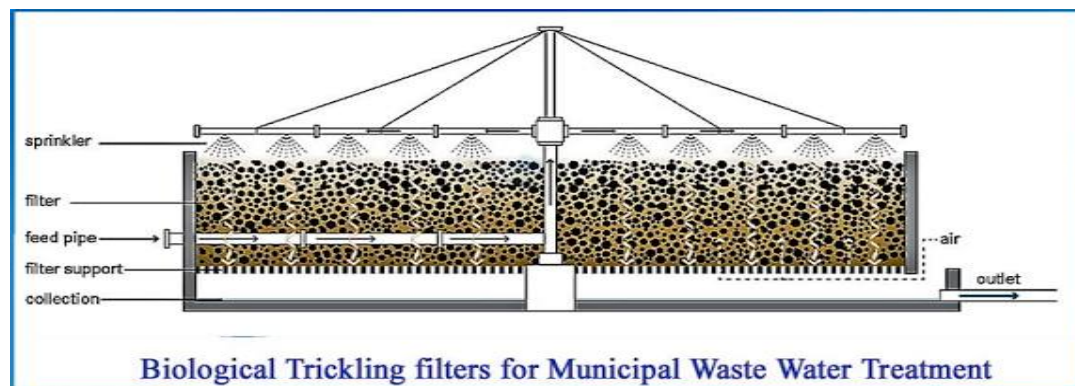
- It is a type of biological treatment process that utilizes microorganisms to break down organic matter in wastewater and remove organic matter from wastewater.

Biological process involved in trickling filter:

Trickling filter is an attached-growth aerobic process in which microorganisms attached to a medium are used for removing organic matter from the wastewater. The biological process in a trickling filter is aerobic oxidation of organic matter by microorganisms attached to a biofilm on the filter media.

Process:

- i. Sewage is distributed on the trickling filter from the top portion of the vessel.
- ii. Air is circulated from the natural drafters or blowers which provides the oxygen for microorganisms attached to the biofilm.
- iii. Organic matter present in the sewage is oxidized by the microorganism attached to the biofilm.
- iv. The sloughed film and the treated wastewater are collected from underdrain.

**Merits of trickling filter in sewage management:**

- i. Simple operation:
Trickling filter can be used for small to medium sized communities and requires less maintenance and operational oversight. The operation cost of the trickling filter is also low.
- ii. High efficiency of sewage treatment:
It is used for the treatment of high concentration of organic matter and helps in removal of 75-90% BOD and suspended solids.
- iii. Low consumption of energy:
The filter uses 30-50% less power than other aerobic treatment processes. These filters produce less sludge than the suspended growth process.
- iv. Flexible rate of loading:
The filter can operate at varying organic and hydraulic loading rates.

Demerits of trickling filter in sewage management:

- i. High cost of capital:
Construction cost is high due to expensive materials and complex infrastructure.
- ii. Odor nuisance:
Standard trickling filters produce fly nuisance and odor nuisance, especially problematic in urban areas.
- iii. Clogging problem:
Frequent clogging of trickling filter from excess biomass accumulation, low performance.
- iv. Requirement of skilled personnel:
Trickling filter requires regular operator and skilled technical personnel for proper management.